Solution Requirements

Date	15 March 2026
Team ID	-
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviours, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviours (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S. No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	User Input	The system shall allow users to upload audio files in supported formats (WAV, MP3, OGG, M4A, WEBM).	High
2	Topic Selection	The system shall allow users to select a reference concept before analysis.	High

3	Speech-to-Text	The system shall convert uploaded speech into text using Faster Whisper.	High
4	Semantic Analysis	The system shall compare the transcript with the reference concept using Sentence-BERT.	High
5	Keyword Matching	The system shall identify matched and missing keywords from the student's explanation.	High
6	Report Generation	The system shall generate a downloadable PDF report containing analysis results.	Medium
7	Dashboard Analytics	The system shall display previous analysis results and performance statistics.	Medium
8	*Database Management*	The system shall store analysis history and reports in an SQLite database.	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Audio analysis should complete within a reasonable time.	Analysis completed within **10–20 seconds** for typical audio files.
2	Scalability	The system should support additional academic topics and future users.	Easily extendable with new reference concepts and modules
3	Security & Data Privacy	Uploaded audio files and analysis results should be handled securely.	User data stored locally in SQLite and temporary files removed after processing.
4	Reliability & Availability	The application should provide consistent results without unexpected failures.	Successfully process supported audio files with minimal errors.
5	Usability & Accessibility	The interface should be simple and user-friendly for students.	Responsive Stream lit UI with clear navigation and downloadable reports.
6	Maintainability	The project should be modular and easy to update.	Separate modules for transcription, semantic analysis, scoring, feedback, and reporting